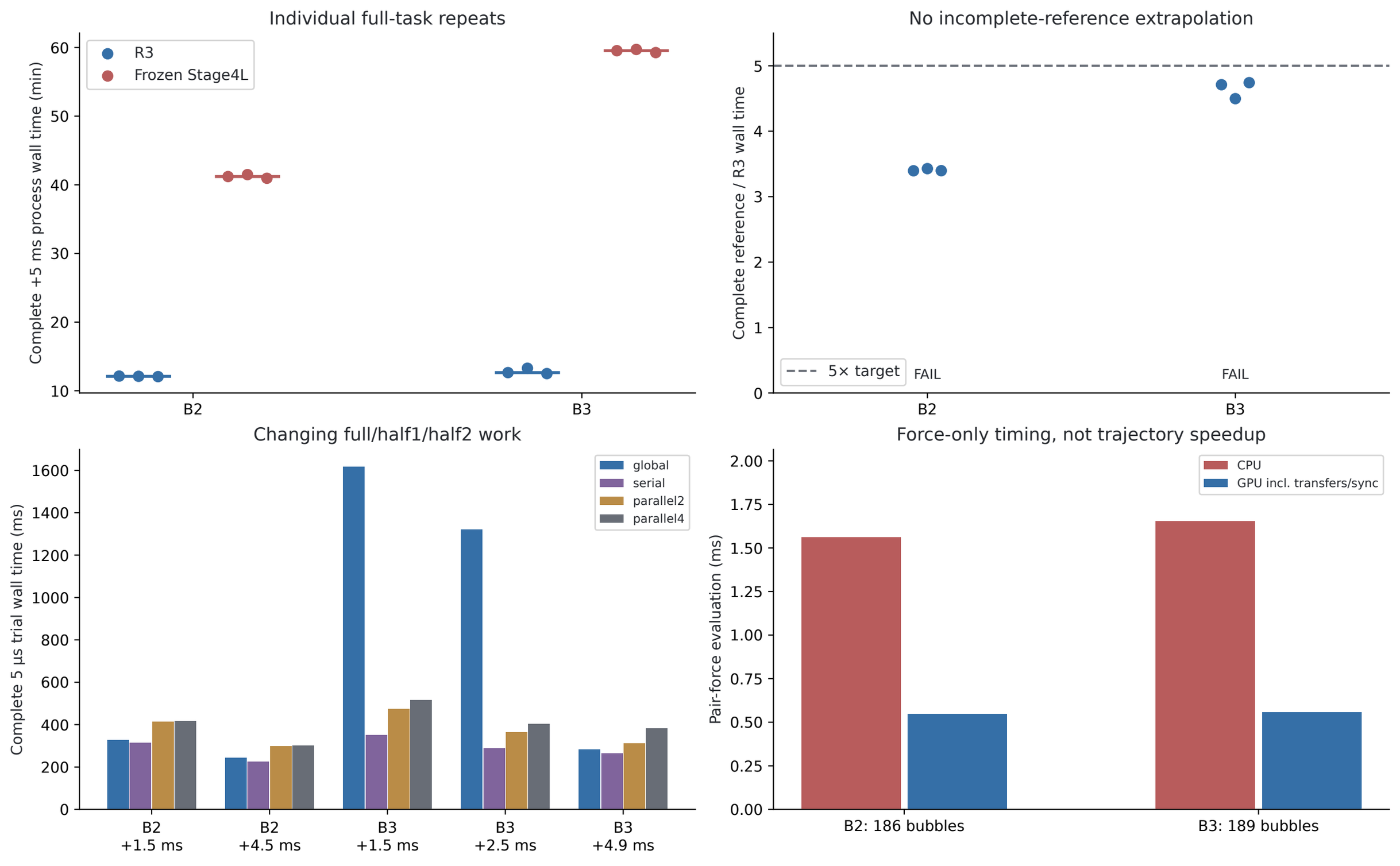
Individual full-task repeats
R3
Frozen Stage4L
Complete +5 ms process wall time (min)
60
50
40
30
20
10
B2
B3
No incomplete-reference extrapolation
Complete reference / R3 wall time
5
4
3
2
1
0
5× target
FAIL
FAIL
B2
B3
Changing full/half1/half2 work
global
serial
parallel2
parallel4
Complete 5 µs trial wall time (ms)
1600
1400
1200
1000
800
600
400
200
0
B2
+1.5 ms
B2
+4.5 ms
B3
+1.5 ms
B3
+2.5 ms
B3
+4.9 ms
Force-only timing, not trajectory speedup
CPU
GPU incl. transfers/sync
Pair-force evaluation (ms)
2.00
1.75
1.50
1.25
1.00
0.75
0.50
0.25
0.00
B2: 186 bubbles
B3: 189 bubbles